Fundamentals of Solid State Physics

The Reciprocal Lattice
倒易点阵

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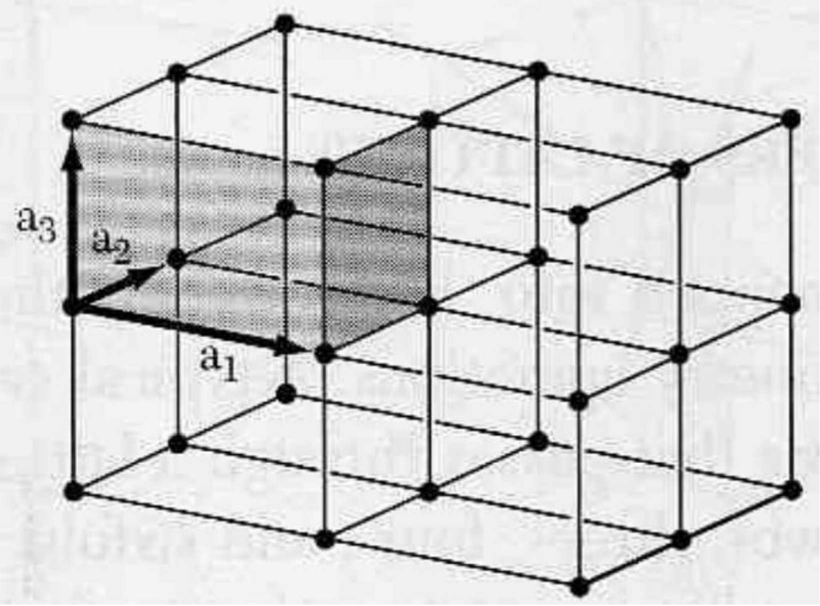
Bravais Lattice 布拉菲点阵

- Each point is *exactly* the same
- Position of each point

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$$

n_1, n_2, n_3 cover *all* the integers (..., −2, −1, 0, +1, +2, ...)

- ($\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$) primitive vectors 基矢量



Real Space
正空间

Direct Lattice
正点阵 / 正格子

Lattice

- **Certain physical properties $F(\mathbf{r})$ in the crystal**
  - **electron density, electrical field, ...**
  - **they can also be periodic**

- **If $F(\mathbf{r})$ is a periodic function**

$$\boxed{F(\mathbf{r}) = F(\mathbf{r}+\mathbf{R})}$$

$$\rightarrow \boxed{F(\mathbf{r}) = \sum_{\mathbf{G}} F_{\mathbf{G}} \exp(i\mathbf{G}\cdot\mathbf{r})}$$

Fourier expansion

$$\rightarrow \boxed{\exp(i\mathbf{G}\cdot\mathbf{R}) = 1}$$

Reciprocal Lattice 倒易点阵

$$\exp(i\mathbf{G}\cdot\mathbf{R})=1$$

- **For a Bravais lattice**

$$\mathbf{R}=n_1\mathbf{a}_1+n_2\mathbf{a}_2+n_3\mathbf{a}_3$$

n_1, n_2, n_3 are integers

- **We define vector G as**

$$\mathbf{G}=m_1\mathbf{b}_1+m_2\mathbf{b}_2+m_3\mathbf{b}_3$$

m_1, m_2, m_3 are integers

(b_1, b_2, b_3) forms reciprocal lattice (倒易点阵 / 倒格子)
G is in the reciprocal space (倒易空间 / 倒空间)

Reciprocal Lattice 倒易点阵

$$\boxed{\exp(i\mathbf{G}\cdot\mathbf{R})=1} \rightarrow \boxed{\mathbf{G}\cdot\mathbf{R}=2\pi\cdot N}$$

- **One solution**

$$\mathbf{b}_1 = 2\pi\frac{\mathbf{a}_2\times\mathbf{a}_3}{\mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)} = 2\pi\frac{\mathbf{a}_2\times\mathbf{a}_3}{V_{\mathbf{R}}}$$

$$\mathbf{b}_2 = 2\pi\frac{\mathbf{a}_3\times\mathbf{a}_1}{\mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)} = 2\pi\frac{\mathbf{a}_3\times\mathbf{a}_1}{V_{\mathbf{R}}}$$

$$\mathbf{b}_3 = 2\pi\frac{\mathbf{a}_1\times\mathbf{a}_2}{\mathbf{a}_1\cdot(\mathbf{a}_2\times\mathbf{a}_3)} = 2\pi\frac{\mathbf{a}_1\times\mathbf{a}_2}{V_{\mathbf{R}}}$$

$$\mathbf{b}_1 \parallel \mathbf{a}_2\times\mathbf{a}_3$$

$$\mathbf{b}_2 \parallel \mathbf{a}_3\times\mathbf{a}_1$$

$$\mathbf{b}_3 \parallel \mathbf{a}_1\times\mathbf{a}_2$$

Reciprocal Lattice 倒易点阵

$$\boxed{\exp(i\mathbf{G}\cdot\mathbf{R})=1} \rightarrow \boxed{\mathbf{G}\cdot\mathbf{R}=2\pi\cdot N}$$

- One can have

$$\boxed{\mathbf{b}_i\cdot\mathbf{a}_j=2\pi\delta_{ij}} \qquad \begin{cases}\delta_{ij}=0, & i\neq j\\ \delta_{ij}=1, & i=j\end{cases}$$

(b_1, b_2, b_3) is primitive vectors to form reciprocal lattice (also a Bravais lattice)

The reciprocal lattice is the Fourier transform of the direct lattice

Reciprocal Lattice 倒易点阵

- **The reciprocal lattice of a Bravais lattice is also a Bravais lattice**

- **The reciprocal lattice of a reciprocal lattice is the original lattice**

- **The primitive cell volume of the reciprocal lattice**

$$V_{\mathbf{G}} = \mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3) = \frac{(2\pi)^3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)} = \frac{(2\pi)^3}{V_{\mathbf{R}}}$$

***V_{R}* is the volume of the original cell, which is a primitive cell**

Reciprocal Lattice 倒易点阵

- 1D lattice

- 2D lattice

- Simple Cubic (SC)

- BCC

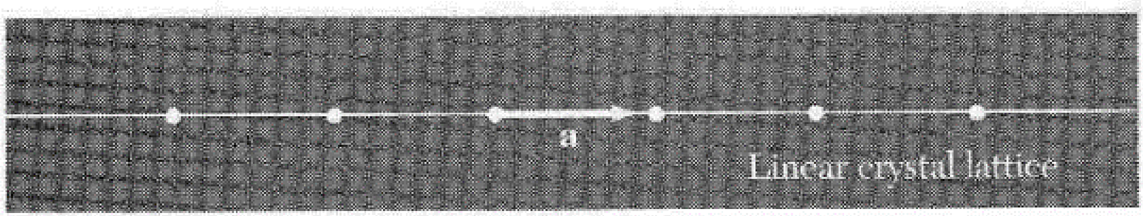
- FCC

1D Lattice

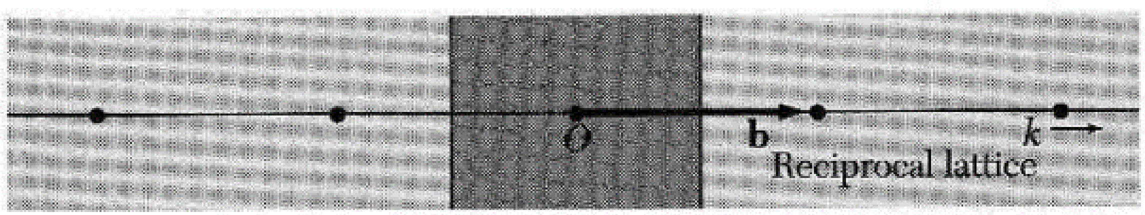
$$\boxed{\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}}$$

$$\boxed{\mathbf{a} = a\hat{\mathbf{x}}}$$

$$\rightarrow \quad \boxed{\mathbf{b} = \frac{2\pi}{a}\hat{\mathbf{x}}}$$



real space



reciprocal space

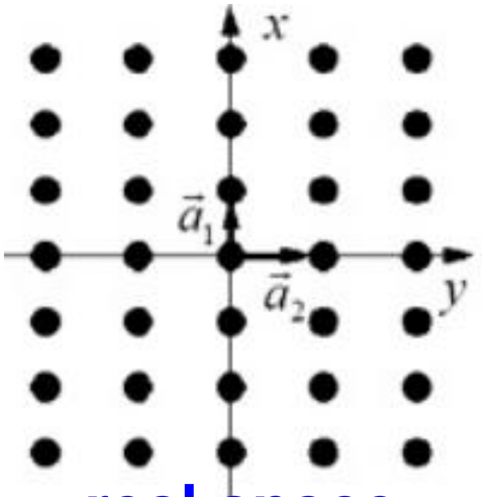
2D Rectangular Lattice

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$$

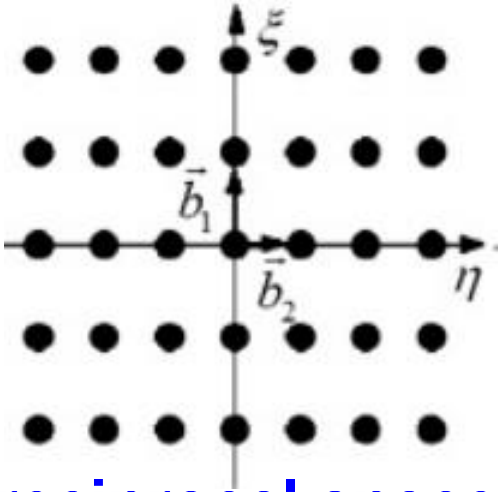
$$\mathbf{a}_1 = a_1\hat{\mathbf{x}}$$
$$\mathbf{a}_2 = a_2\hat{\mathbf{y}}$$

$\rightarrow$

$$\mathbf{b}_1 = \frac{2\pi}{a_1}\hat{\mathbf{x}}$$
$$\mathbf{b}_2 = \frac{2\pi}{a_2}\hat{\mathbf{y}}$$



$\rightarrow$



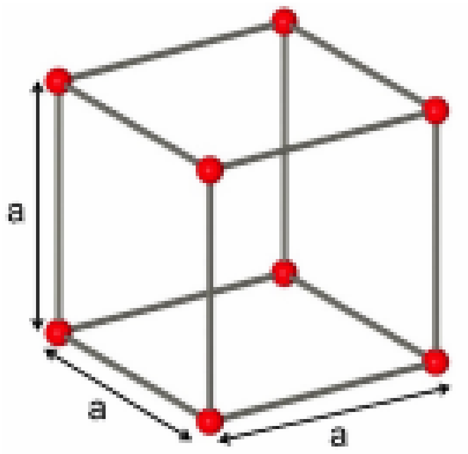
real space

reciprocal space

Simple Cubic (SC)

primitive vectors

$$\mathbf{a}_1 = a\hat{\mathbf{x}}$$
$$\mathbf{a}_2 = a\hat{\mathbf{y}}$$
$$\mathbf{a}_3 = a\hat{\mathbf{z}}$$



direct lattice

$$\mathbf{b}_1 = \frac{2\pi}{a}\hat{\mathbf{x}}$$
$$\mathbf{b}_2 = \frac{2\pi}{a}\hat{\mathbf{y}}$$
$$\mathbf{b}_3 = \frac{2\pi}{a}\hat{\mathbf{z}}$$

the reciprocal lattice
is still SC

BCC

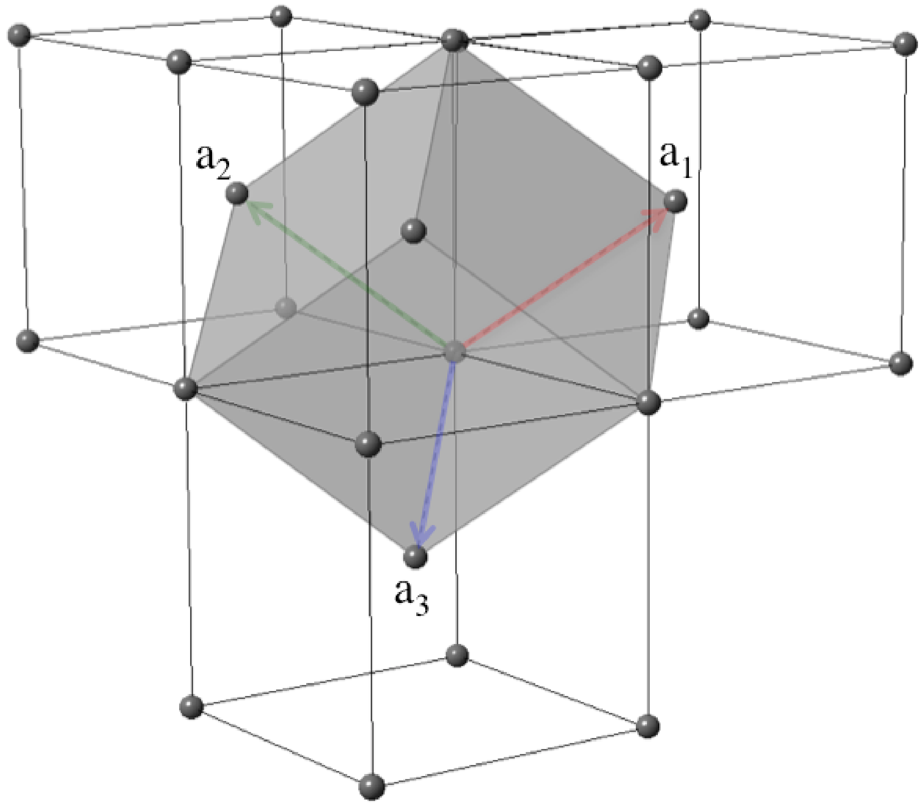
primitive vectors

$$\mathbf{a}_1 = \frac{a}{2}\left(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}\right)$$

$$\mathbf{a}_2 = \frac{a}{2}\left(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}\right)$$

$$\mathbf{a}_3 = \frac{a}{2}\left(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}\right)$$

primitive cell



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$$\mathbf{a}_3 = \frac{a}{2}\left(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}\right)$$

$$\rightarrow$$

$$\mathbf{b}_1 = \frac{4\pi}{a}\frac{1}{2}\left(\hat{\mathbf{y}} + \hat{\mathbf{z}}\right)$$

$$\mathbf{b}_2 = \frac{4\pi}{a}\frac{1}{2}\left(\hat{\mathbf{z}} + \hat{\mathbf{x}}\right)$$

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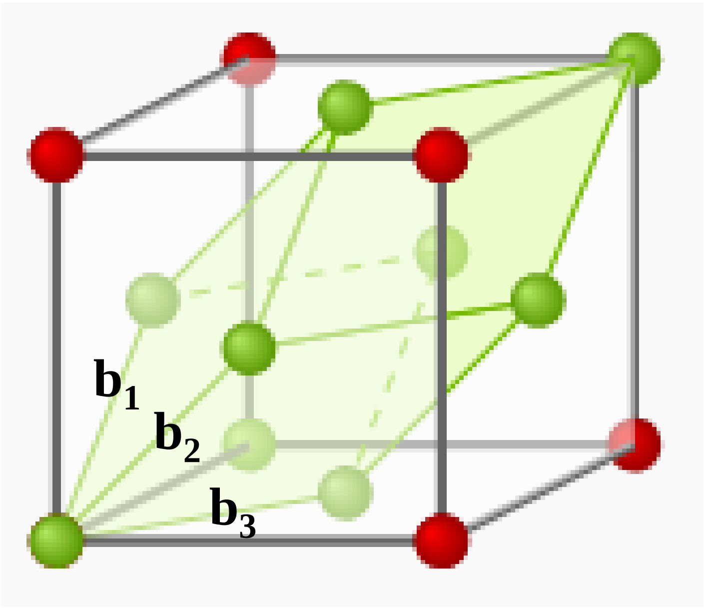
BCC



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The reciprocal lattice of BCC is FCC

FCC

primitive vectors

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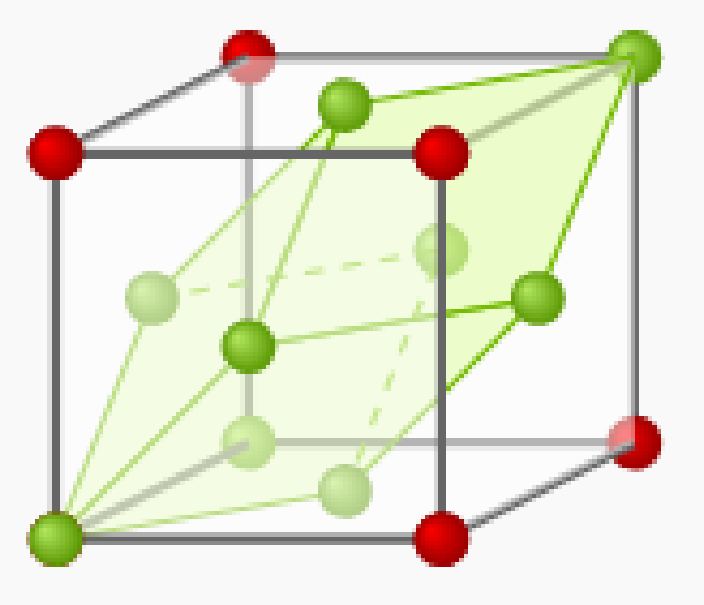
primitive cell



FCC

primitive vectors

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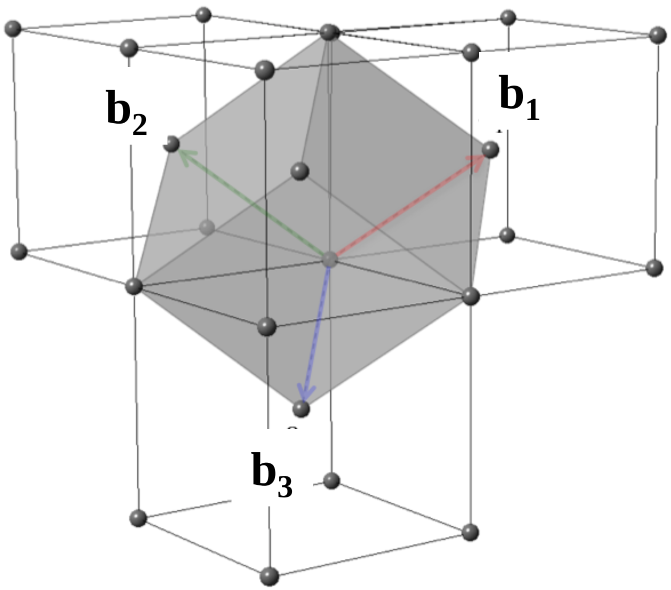
FCC



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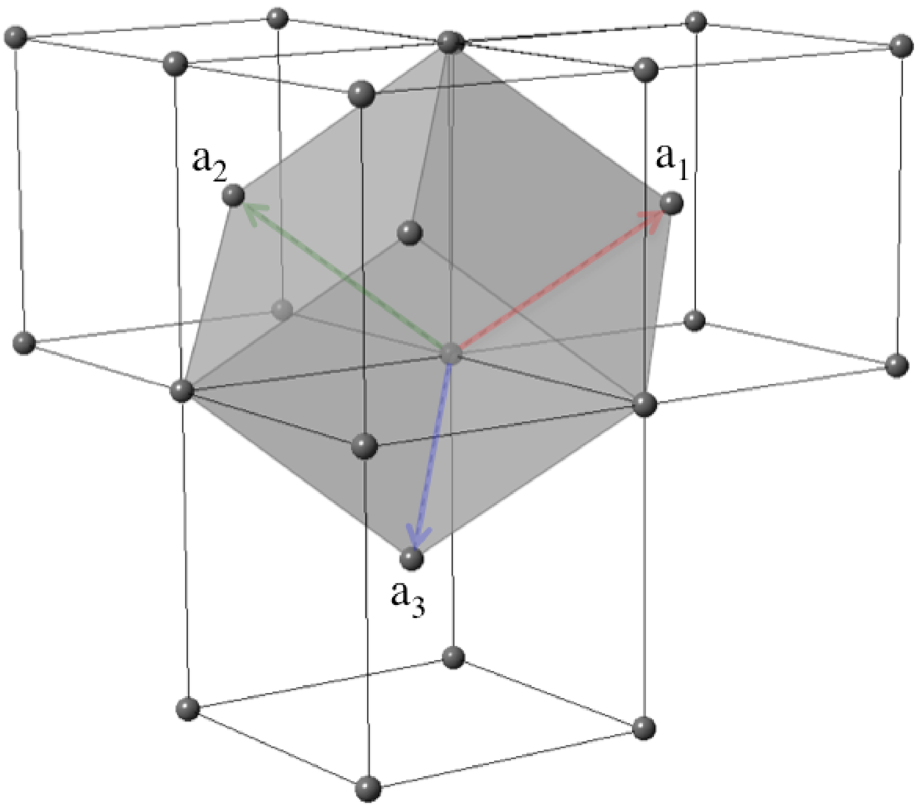
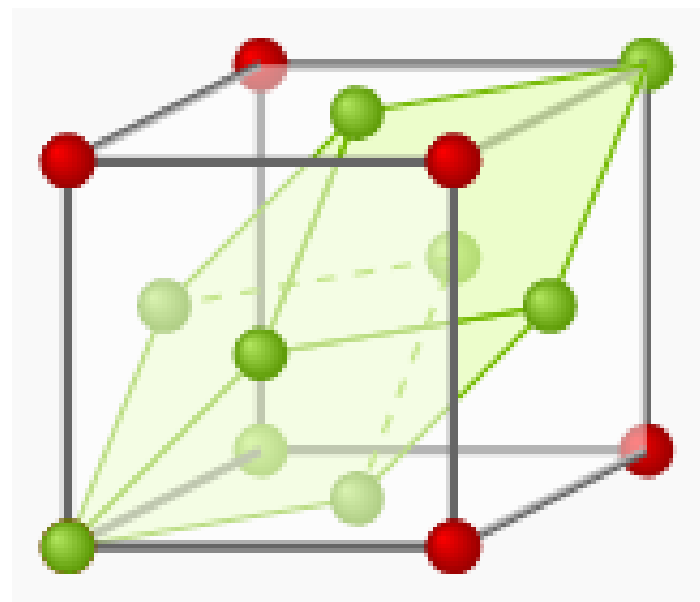
The reciprocal lattice of FCC is BCC

BCC and FCC

BCC



FCC

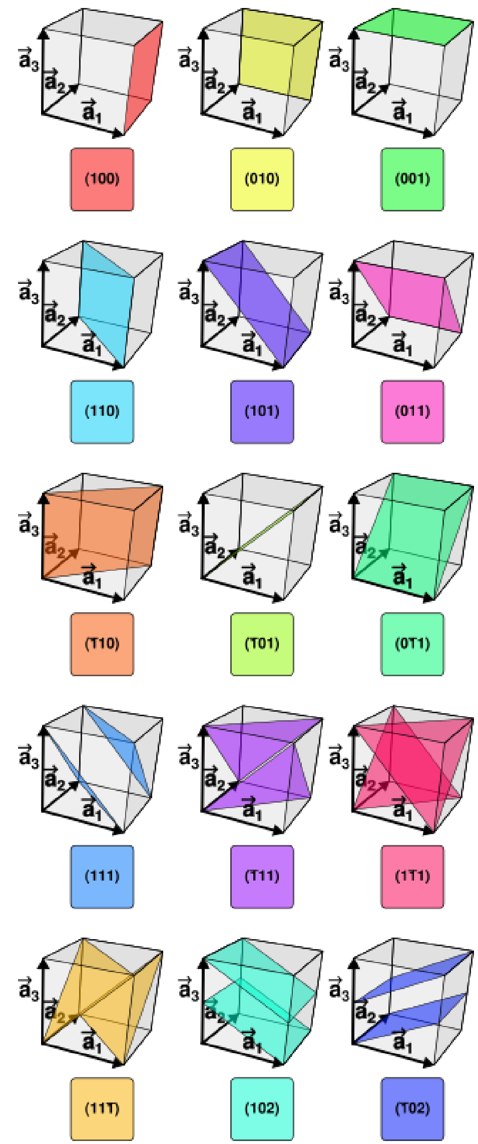


The reciprocal lattice of BCC is FCC

The reciprocal lattice of FCC is BCC

Miller Indices of Crystal Planes 密勒指数

- **crystal plane (hkl)**
  - intercepts at (a_1/h, a_2/k, a_3/l)
  - hkl are integers with no common factors



For all Bravais lattices, not only cubic we have:

1. The (hkl) plane ⊥ reciprocal lattice vector G

$$\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$$

2. The interplanar distance of (hkl) plane $d_{(hkl)}$

$$d_{(hkl)} = \frac{2\pi}{|\mathbf{G}|}$$

Brillouin Zones 布里渊区

- The *First Brillouin Zone (FBZ)*
  - the Wigner-Seitz cell of the reciprocal lattice

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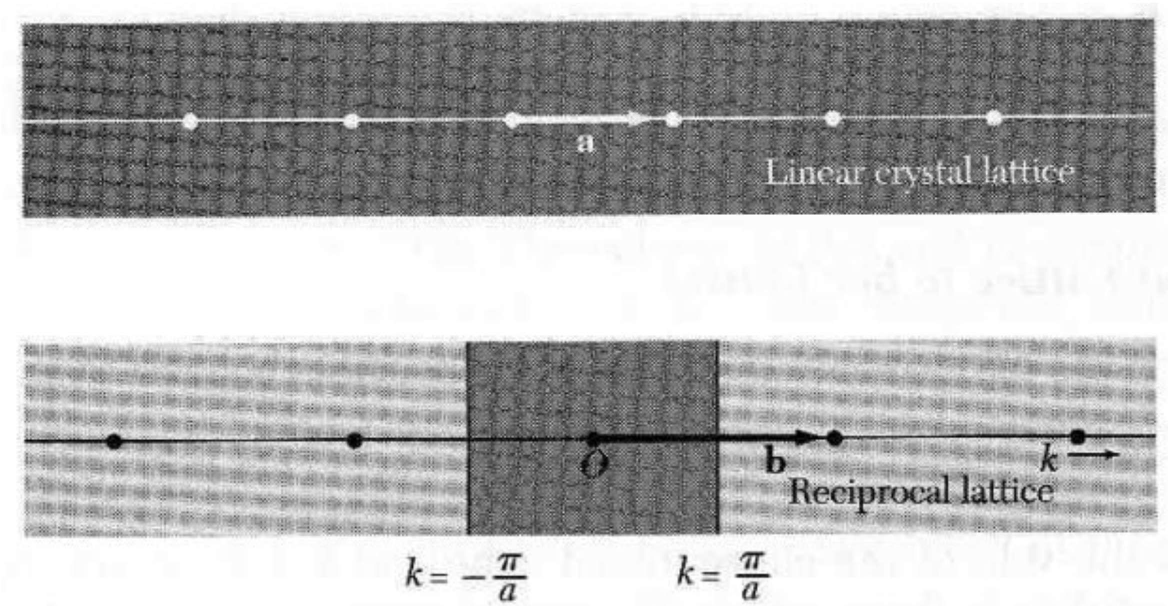

real space

reciprocal space

1D lattice, FBZ

$$-\frac{\pi}{a} < k < \frac{\pi}{a}$$

Brillouin Zones 布里渊区

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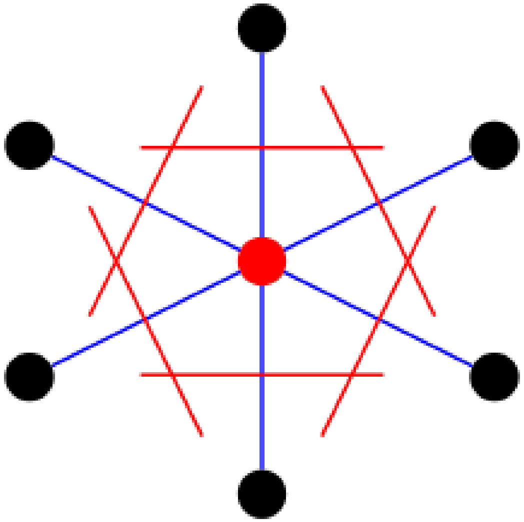

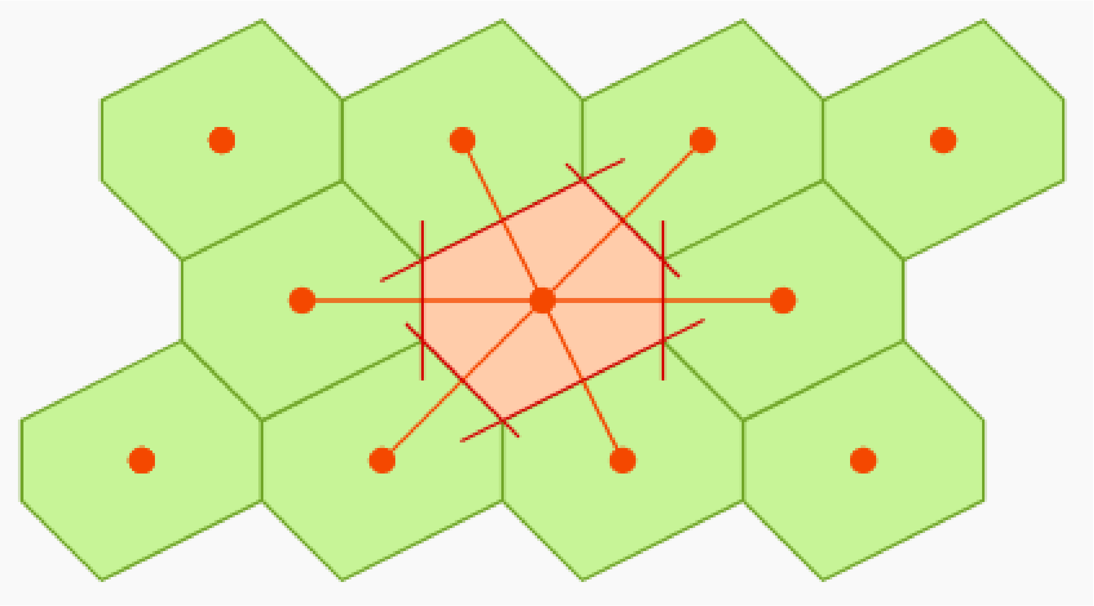

2D lattice

Brillouin Zones 布里渊区

- The *First Brillouin Zone (FBZ)*
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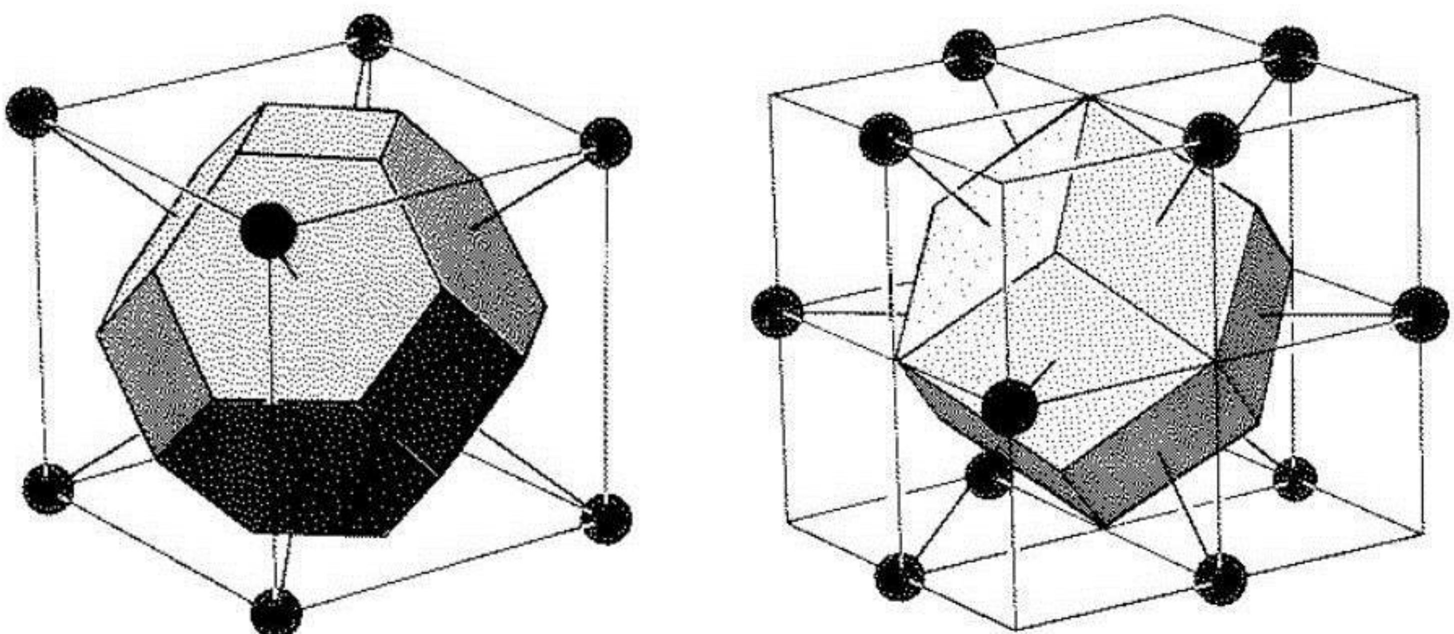

3D lattice

BCC

FCC

Wigner-Seitz Cell

Brillouin Zones 布里渊区

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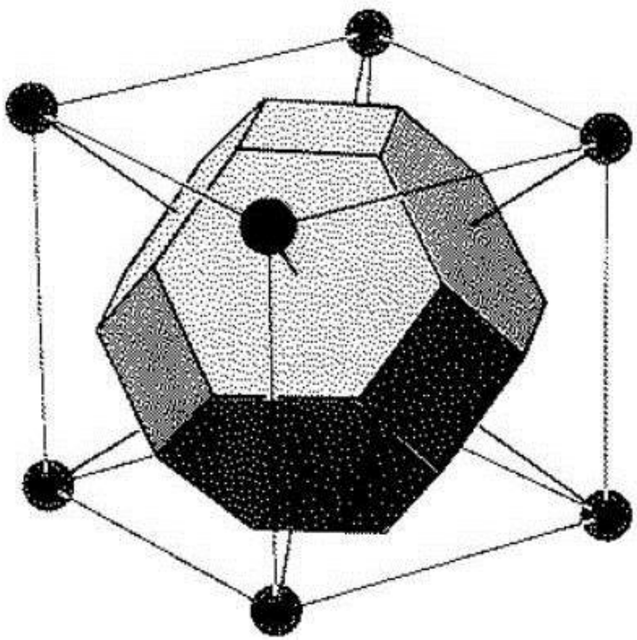
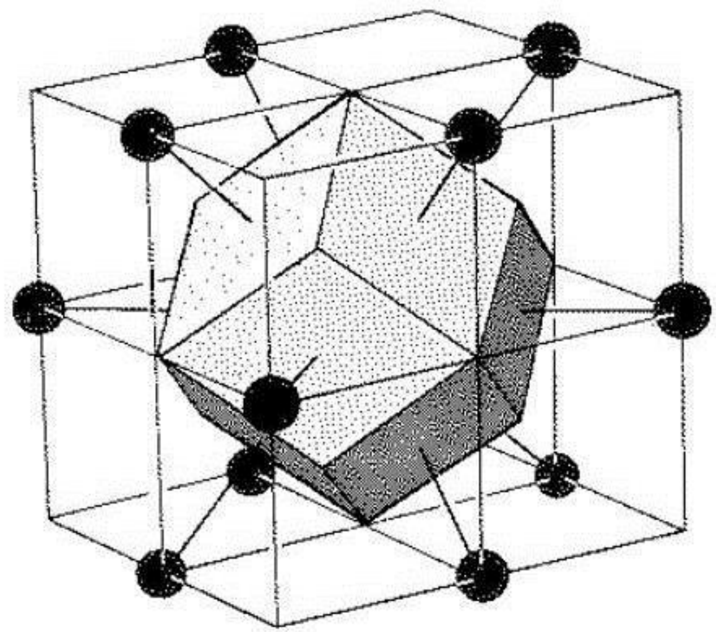


FBZ for FCC

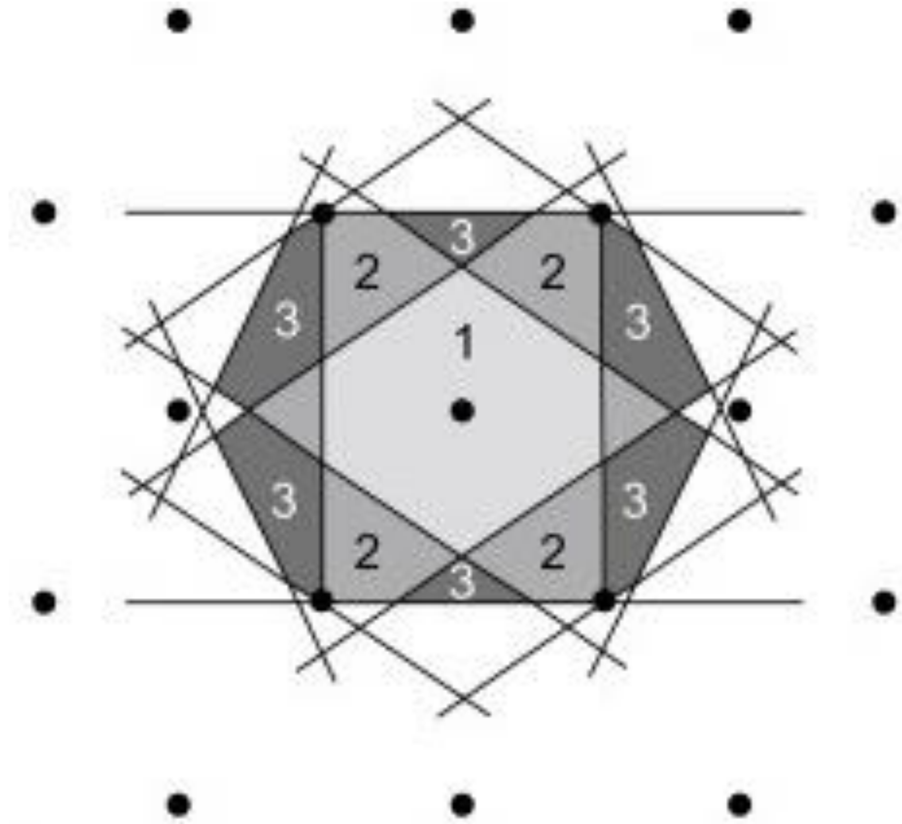


FBZ for BCC

Q: What is the volume of the FBZ?

Brillouin Zones 布里渊区

- **Higher-order Brillouin Zones (BZs)**
  - **2nd BZ**
  - **3rd BZ**
  - **...**



All the Brillouin Zones have the same area / volume.

Brillouin Zones 布里渊区

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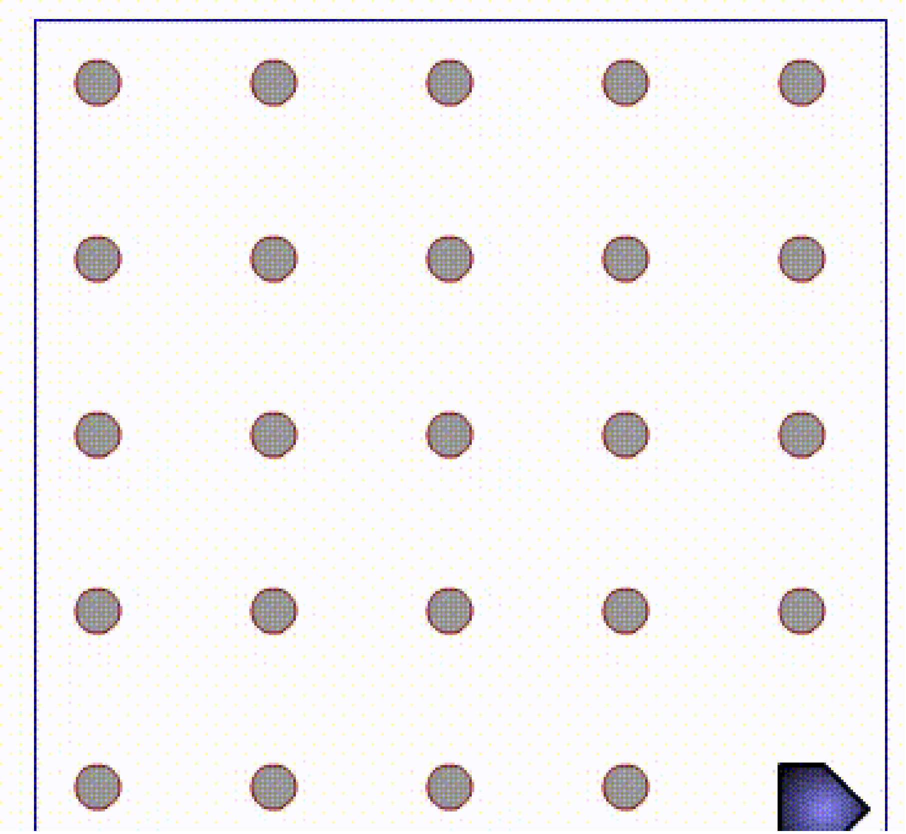
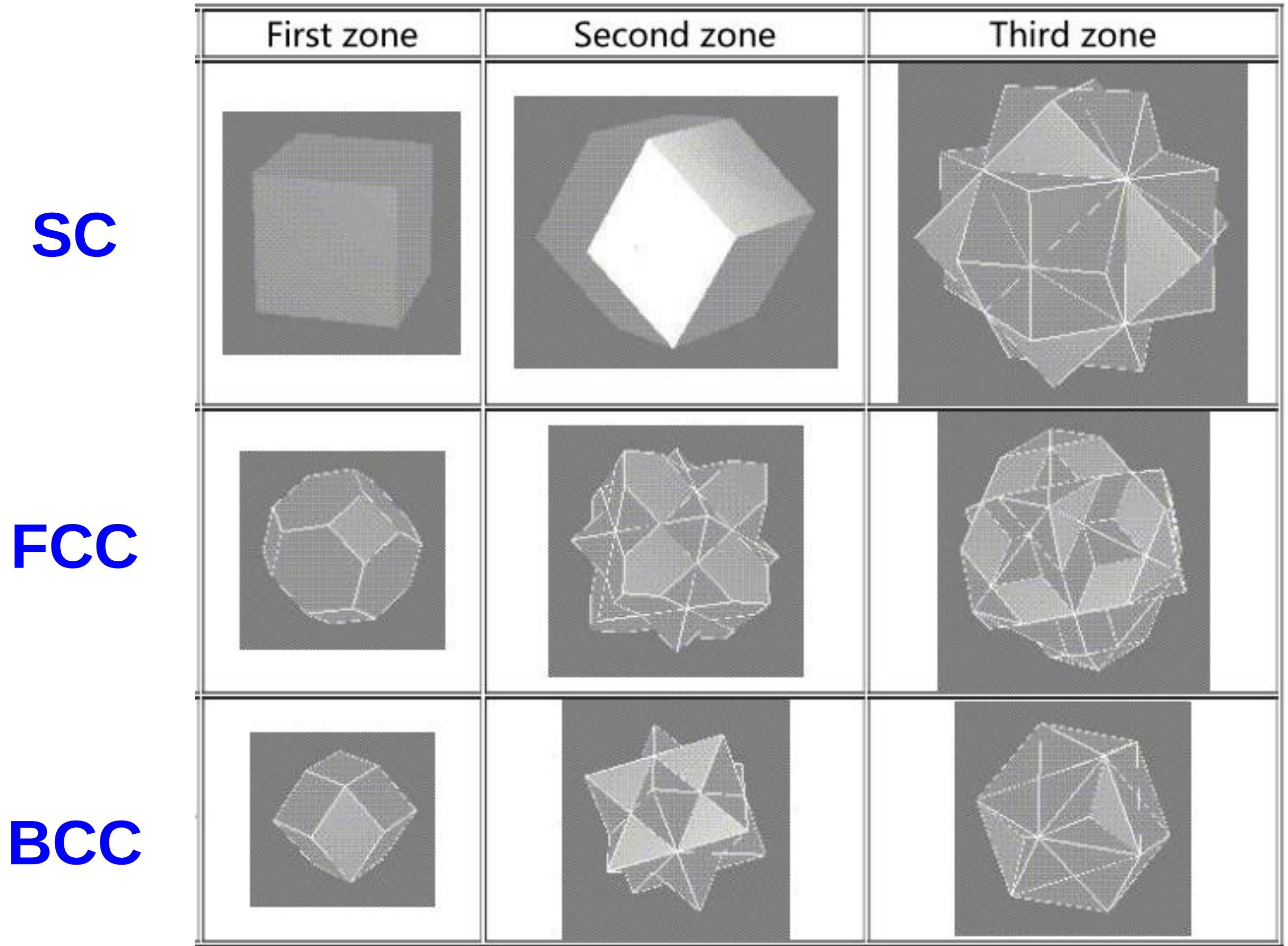


Brillouin Zones 布里渊区



All the Brillouin Zones have the same volume.

Thank you for your attention